RAMS for Railway Systems

Subir Chowdhury School of Quality and Reliability

RAMS Terms and Definitions

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Introduction: Subir Chowdhury School of Quality and Reliability

Formerly known as:

"Reliability Engineering Centre"

Introduction

- Came into existence in 1981 with financial support of more than 10 million from MHRD, Govt. of India.
  - To create an awareness about Reliability Engineering Education,
  - Research and its applications in all aspects of Engineering & Technology.
- Interdisciplinary M. Tech. Program started from the academic year 1982-83.

Introduction ...

- Programs primarily are at
  - Post graduate level. However, some elective courses are being offered to final year UG students.
  - Guidance and facility for research leading to Ph. D. Degree.
- Inputs to the program includes fresh graduates through GATE and sponsored candidates from
  - Defense Establishments.
  - Private/public sector Industries
  - Academic Institutions
- So far more than 98% employment to our students.

Faculty

- Prof. S. Mukhopadhyay, Chairperson
- Prof. V.N.A. Naikan
- Prof. S.K. Chaturvedi
- Dr. N.K. Goyal
- Dr. Monalisa Sarma
- Dr. R. N. Rai
- Dr. Heeralal Gargama
- Dr. Indranil Hazra

Facilities

- Environment Test Chambers to carry out a few environmental tests.
- The school is equipped with the sophisticated instruments, test fixtures and latest computer packages for reliability related simulation and analysis of component/product/system behavior.
- Library books and International/National Journals to impart Reliability education to the students, researchers and practitioners.
- In contact with some other organizations which can provide testing facilities as per requirements.

Industrial Research Applications

- Environmental Testing of components, assemblies
- Failure Analysis of Engineering Components and Systems
- Probabilistic Risk/ Safety Analysis
- Software Reliability Modeling & Testing
- Condition Monitoring and Fault Diagnosis
- System RAMS Modeling and Analysis

FACULTY INTRODUCTION

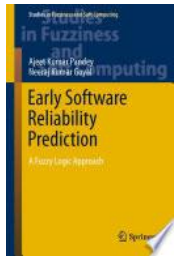
Dr. Neeraj Kumar Goyal

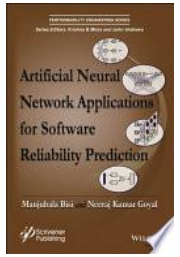
Brief Biography

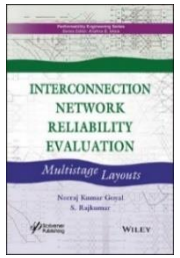
- Serving as Professor in Subir Chowdhury School of Quality and Reliability (SCSQR), IIT Kharagpur.
  - Also Prof. Incharge of Ranking Cell, IIT Kharagpur;
  - BIS (LITD02) review committee member.
  - Managing Editor, Life Cycle Reliability and Safety Engineering
  - Managing Committee Member, SRESA
- PhD in Reliability Engineering from IIT Kharagpur in year 2006 and since then working as faculty at SCSQR, IIT Kharagpur.
- Completed BE in Electronics and Communication from MNIT Jaipur in year 2000
- Worked as an executive for one year in Secure Meters Ltd., Udaipur till 2001.
- Continuously engaged in industry-sponsored research and consultancy projects from various organizations:
  - DRDO, MTRDC, DIC, ISRO, NPCIL, BARC, Secure Meters, Vodaphone, AERB, Ministry of Railways, MHRD, DST, Indian Army, IGCAR, John Deere, CESC, Kimbal, L&T etc.
- Research Guidance
  - PhD: 8 Completed and 5 Ongoing
  - MS: 2 Completed

Publications

- Books:
  - Early Software Reliability Prediction: A fuzzy logic approach
    - Authors: Ajeet Kumar Pandey and N.K. Goyal
    - Publisher: Springer, 2013
  - Artificial Neural Network Applications for Software Reliability Prediction
    - Authors: Manjubala Bisi and N.K. Goyal
    - Publisher: Scrivener Publishing, WILEY, 2017
  - Interconnection Network Reliability Evaluation
    - Authors: Neeraj Kumar Goyal, S. Rajkumar
    - Publisher: Scrivener Publishing, WILEY, 2020
- Papers
  - International Journals: 50
  - International Conferences: 22







Consultancy Projects

#	Project Name	Sponsored By	Period
1	Reliability Work Package of LRSAM	DRDL, Hyderabad	2007-08
2	RAMS Model for Project ASTRA	DRDL, Hyderabad	2007-08
3	Shutdown PSA of KAPS	NPCIL, Mumbai	2007-08
4	Accelerated Life Testing of 30-pin Connector	BARC, Mumbai	2008-09
5	Flood PSA of KAPS	NPCIL, Mumbai	2009-11
6	Assessment of Residual Reliability of Armored Fighting Vehicles through CBM	MCEME, Hyderabad	2010-11
7	Advance Studies on Human Reliability Analysis	NPCIL, Mumbai	2012-13
8	Reliability Modeling and Prediction of Process Control System	DRDO, Panagarh	2012-13
9	Reliability Assessment and Improvement	Secure Meters, Udaipur	2014-15
10	Preliminary Risk Analysis of Hypersonic Technology Demonstrator Vehicle	DRDL, Hyderabad	2014-15
11	Flood (Internal and External) Probabilistic Safety Assessment of MAPS	MAPS, NPCIL, Madras	2015-16
12	Application of RAMS concepts for HHP (4500HP) Diesel Locomotives	DLW, Varanasi	2015-16
13	Reliability Assessment and Reliability Improvement of Products Designed	Secure Meters, Udaipur	2015-16
14	Reliability Work Package for Project LR-SAM	DRDL, Hyderabad	2015-16
15	Reliability Assessment and Reliability Improvement of Products Designed	Secure Meters, Udaipur	2016-17

Consultancy Projects…(2)

#	Project Name	Sponsored By	Period
16	Reliability Studies on Electronic Power Conditioning (EPC) Of Microwave Power Module	MTRDC, DRDO, Bengaluru	2015-16
17	Standard Based Reliability, Prediction of Submarine Degaussing System	L&T Ltd., Mumbai	2017-18
18	Software Quality and Reliability	John Deere, Pune	2020-21
19	Reliability Study of RUSTAM-II	ADE, Bangalore	2020-21
20	Resiliency Modeling and Analysis with Recommendations for Overhead LV Distribution Network of CESC Ltd. in Kolkata Region	CESC, Kolkata	2021-22
21	Reliability Assessment and Reliability Improvement of Products Designed	Kimbal, Delhi	2024-25
22	Review of Product Reliability Modeling, Prediction, and Analysis Reports	L&T Defence, Mumbai	2024-24

Research Projects

#	Project Name	Sponsored By	Period
1	Design of Minimal Cost Backbone Network Layout for Given Capacity and Reliability Requirements	VICET, IIT Kharagpur	2009-13
2	Reliability Assessment of the Large Complex Computer Code	AERB, Mumbai	2009-11
3	Reliability Assessment of L40 Stage Assembly and Integration Process	ISRO, Bengaluru	2016-19
4	Reliability Studies on Electronics Power Conditioning (EPC) of Microwave Power Module	MTRDC, DRDO, Bengaluru	2015-16
5	Developing a Reliability Engineering Framework for Indian Railways Rolling Stock	DST	2016-20
6	Reliability Studies on Electronics Power Conditioning (EPC) for Space Application	MTRDC, DRDO, Bengaluru	2016-17
7	Quantification of Software Reliability for Computer-Based I & C Systems of Prototype Fast Breeder Reactor (PFBR)	Indira Gandhi Centre For Atomic Research (IGCAR), Kalpakkam	2017-18
8	Design, Development, Verification and Reliability Analysis of Prototype Control Room HMI Software for Fast Breeder Reactors	Indira Gandhi Centre For Atomic Research (IGCAR), Kalpakkam	2019-22
9	INTPART: Collaborative Research Based Education for Optimized Performance of Wind Farms	Western Norway University of Applied Sciences (HVL)	2023-28
10	AI Powered Public Battery Swapping Station for e-rickshaws	IIT KGP AI4ICPS HUB	2024-26

Short Term Courses Organized

Course Title	Category	Role	Start Date	End Date	Company
Probabilistic Safety Assessment	GIAN	Principal Coordinator	07-12-2015	18-12-2015	
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	27-07-2015	31-07-2015	
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	11-07-2016	15-07-2016	
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	10-07-2017	14-07-2017	
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	09-07-2018	13-07-2018	
Reliability Engineering - Module 2	Sponsored	Co-Coordinator	19-11-2018	22-11-2018	TVS
Reliability Engineering Module-3	Sponsored	Co-Coordinator	25-02-2019	28-02-2019	TVS
Itroduction to Software Rreliability and Life Testing	Sponsored	Principal Coordinator	30-09-2019	01-10-2019	John Deere
RAMS for Railway Systems	Sponsored	Principal Coordinator	21-10-2019	23-10-2019	Siemens
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	08-07-2019	12-07-2019	
Reliability Engineering for Engineers of Whirlpool Corporation	Sponsored	Principal Coordinator	11-01-2021	24-08-2021	Whirlpool
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	14-12-2020	18-12-2020	
RAMS for Railway Systems	Sponsored	Principal Coordinator	21-11-2022	23-11-2022	CLW
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	05-12-2022	09-12-2022	

Short Term Courses Organized…(2)

RAMS for Railway Systems

Subir Chowdhury School of Quality and Reliability

Course Title	Category	Role	Start Date	End Date	Company
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	06-12-2021	10-12-2021	
RAMS for DMDE	Sponsored	Principal Coordinator	14-12-2021	16-12-2021	DMDE
RAMS for Railway Systems	Self Sponsosred	Principal Coordinator	05-12-2021	9-12-2021	
Module 1: Introduction to Reliability Engineering	Sponsored	Co-Coordinator	24-03-2022	25-03-2022	Titan
Module 2: System Reliability Modeling	Sponsored	Principal Coordinator	04-05-2022	05-05-2022	Titan
Railway RAMS for RITES	Sponsored	Principal Coordinator	14-06-2023	16-06-2023	RITES
RAMS for Railway Systems	Self Sponsored	Principal Coordinator	03-07-2023	07-07-2023	
Probability and Statistics	Sponsored	Co-Coordinator	22-06-2023	23-06-2023	TVS
Reliability Modeling and Analysis	Sponsored	Co-Coordinator	21-08-2023	26-08-2023	TVS
Design for Reliability	Sponsored	Co-Coordinator	6-11-2023	9-11-2023	TVS
Reliability Design Practices	Sponsored	Principal Coordinator	22-09-2023	23-09-2023	Skanray
Reliability Testing	Sponsored	Co-Coordinator	4-03-2024	7-03-2024	TVS
Maintenance Engineering	Sponsored	Co-Coordinator	4-03-2024	7-03-2024	TVS
RAMS for Railway Systems	Self Sponsored	Principal Coordinator	08-07-2024	12-07-2024	

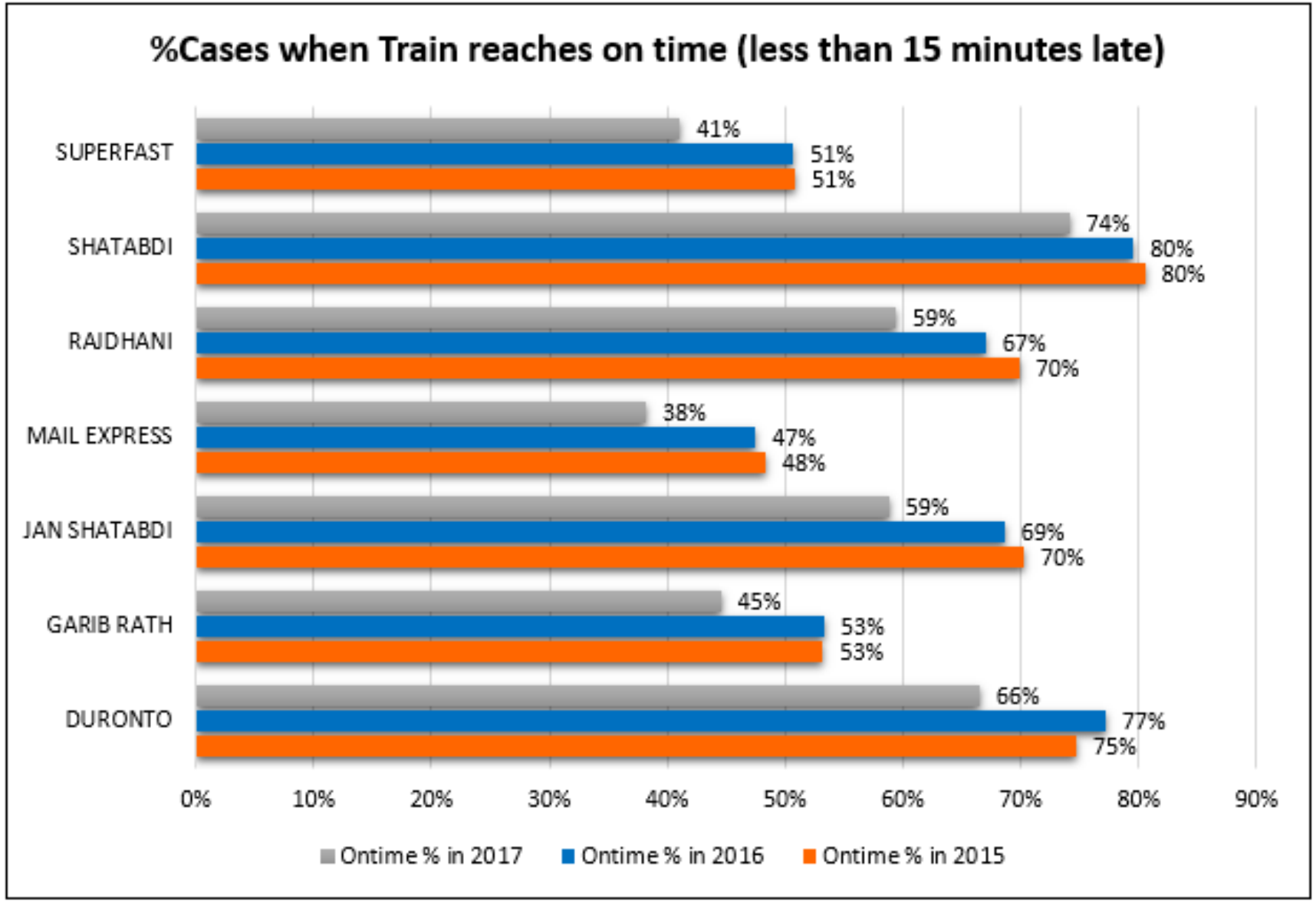
Terms and Definitions

- Reliability
- Availability
- Maintainability
- Safety

Reliability and
Safety???

Whom Should
It Bother?
216493/C
SLEEPER
शयनयान
दप SW

Delayed Trains!!
Is it a real concern???



Do Metro Trains have Different Fate?

How to address these concerns into design, manufacturing and maintenance???

RAMS Measurement and Assurance!!!

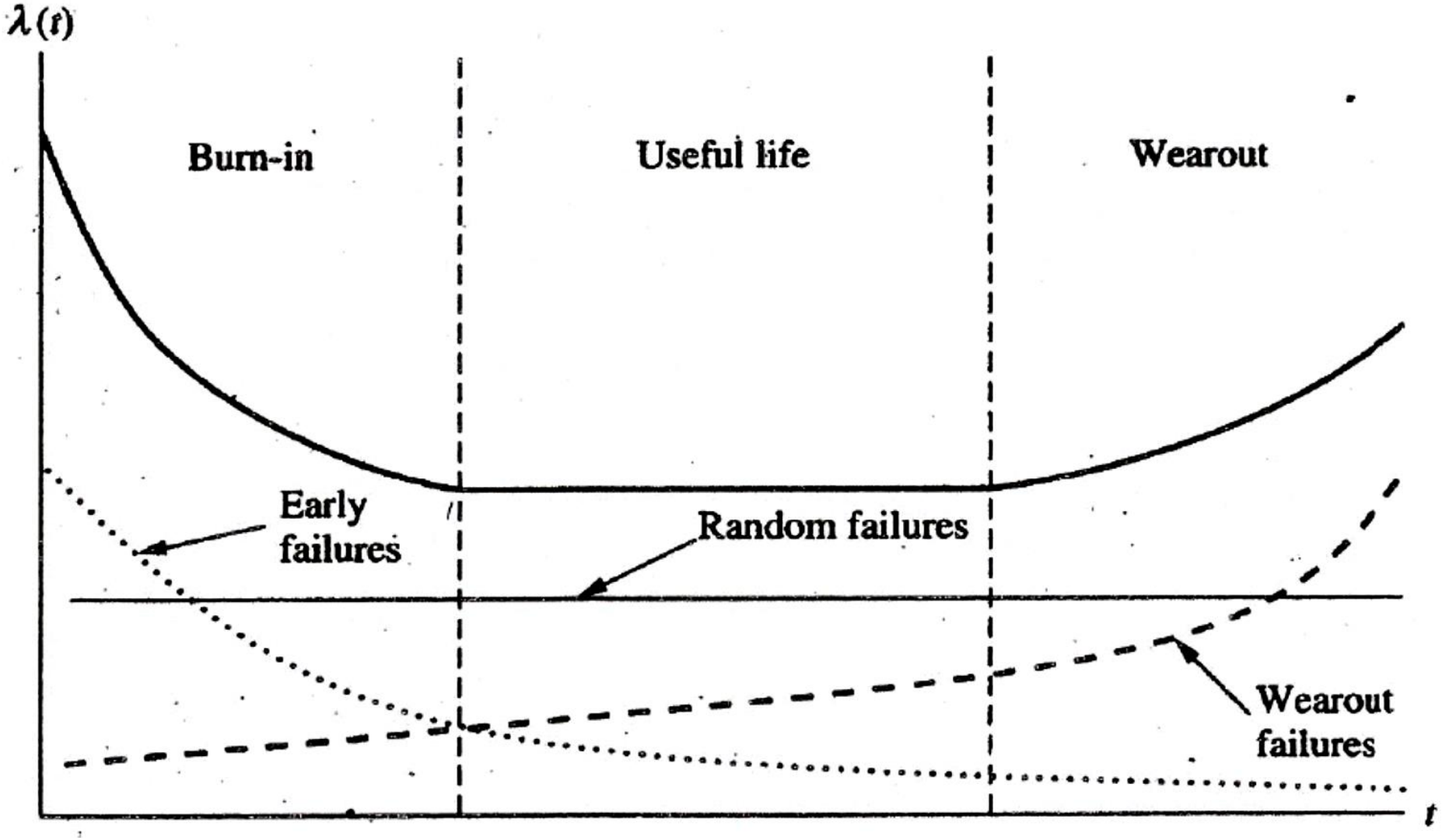
RAMS Definitions

- Reliability
  - Reliability is defined to be probability that a product/system will perform the required function for a given period of time when used under stated operating conditions.
- Availability
  - The probability of a product to be in a state to perform a required function under given conditions at a given instant of time or over a given time interval assuming that the required external resources are provided.
- Maintainability
  - The probability of an item under given conditions of use, to be retained in, or restored to, a state in which it can perform a required function, when maintenance is performed under given conditions and using stated procedures and resources.
- Safety
  - Freedom from unacceptable risk to human health or to the environment.
- Risk
  - The combination of expected frequency of loss and the expected degree of severity of that loss.
- For any product of your choice, explain RAMS terms.
- What are the random variables of your interest?

Reliability Related Terms

- Reliability
  - $R(t) = \Pr\{T \geq t\}$ where is Time To Failure (TTF) random variable.
- Unreliability
  - It is cumulative distribution function (CDF) of TTF distribution.
  - $F(t) = 1 - R(t) = \Pr\{T < t\}$
- PDF of TTF
  - $f(t) = \frac{dF(t)}{dt} = -\frac{dR(t)}{dt}$
- Mean Time to Failure (MTTF)
  - $MTTF = E(T) = \int_0^\infty t f(t) dt = \int_0^\infty R(t) dt$
- Failure Rate or Hazard Rate
  - It provides an instantaneous rate of failure at time t.
  - It is conditional probability of failure per unit time in time interval (t, t+Δt).
  - $\lambda(t) = \lim_{\Delta t \to 0} \frac{R(t) - R(t+\Delta t)}{R(t)\Delta t} = -\frac{dR(t)}{dt} \cdot \frac{1}{R(t)} = \frac{f(t)}{R(t)}$

Bath Tub Curve



Constant Failure Rate (CFR) Model (Exponential Distribution)

- Hazard rate is constant (does not change with time)

$$\lambda(t) = z(t) = \lambda$$

- Reliability

$$R(t) = e^{-\int_0^t \lambda dx} = e^{-\lambda t},$$
$$where\ t \geq 0$$

- Unreliability

$$F(t) = 1 - R(t) = 1 - e^{-\lambda t}$$

- Probability Density Function (PDF)

$$f(t) = \lambda(t)R(t) = \lambda e^{-\lambda t}$$

- Mean Time to Failure

$$\text{MTTF} = \int_0^\infty e^{-\lambda t} dt$$
$$= -\frac{1}{\lambda} e^{-\lambda t} \Big|_0^\infty = \frac{1}{\lambda}$$

- Variance

$$\sigma^2 = \int_0^\infty t^2 f(t)\ dt - MTTF^2$$
$$= MTTF^2 = \frac{1}{\lambda^2}$$

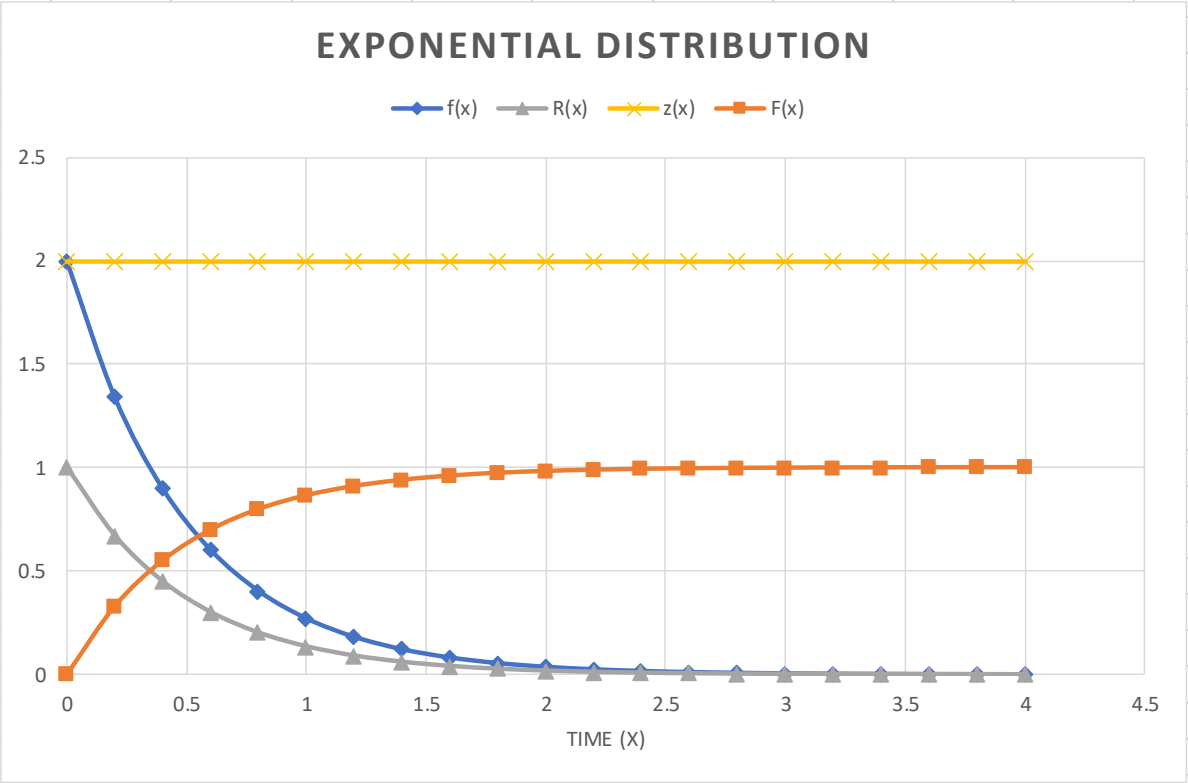
- Standard Deviation

$$\sigma = \frac{1}{\lambda} = \text{MTTF}$$

  – SD is equal to MTTF. This means as MTTF increases, variability also increases.

λ	2		MTTF	0.5

x	f(x)	F(x)	R(x)	z(x)
0	2	0.00	1.00	2.00
0.2	1.340640092	0.33	0.67	2.00
0.4	0.898657928	0.55	0.45	2.00
0.6	0.602388424	0.70	0.30	2.00
0.8	0.403793036	0.80	0.20	2.00
1	0.270670566	0.86	0.14	2.00
1.2	0.181435907	0.91	0.09	2.00
1.4	0.121620125	0.94	0.06	2.00
1.6	0.081524408	0.96	0.04	2.00
1.8	0.054647445	0.97	0.03	2.00
2	0.036631278	0.98	0.02	2.00
2.2	0.02455468	0.99	0.01	2.00
2.4	0.016459494	0.99	0.01	2.00
2.6	0.011033129	0.99	0.01	2.00
2.8	0.007395727	1.00	0.00	2.00
3	0.004957504	1.00	0.00	2.00
3.2	0.003323115	1.00	0.00	2.00
3.4	0.00222755	1.00	0.00	2.00
3.6	0.001493172	1.00	0.00	2.00
3.8	0.001000903	1.00	0.00	2.00
4	0.000670925	1.00	0.000335463	2



Weibull Distribution

- Hazard rate function:

$$\lambda(t) = \left(\frac{\beta}{\theta}\right)\left(\frac{t}{\theta}\right)^{\beta-1}, t \geq 0$$

  – The parameters θ and β are positive values estimated from failure data.
    - Parameter θ is called scale parameter or characteristic life.
    - Parameter β is called shape parameter.

- Reliability

$$R(t) = exp\left[-\int_0^t \frac{\beta}{\theta}\left(\frac{x}{\theta}\right)^{\beta-1} dx\right] = e^{-\left(\frac{t}{\theta}\right)^{\beta}}$$

  – At $t = \theta$, $R(\theta) = 0.368$

- PDF

$$f(t) = -\frac{dR(t)}{dt} = \frac{\beta}{\theta}\left(\frac{t}{\theta}\right)^{\beta-1} e^{-\left(\frac{t}{\theta}\right)^{\beta}}$$

$$MTTF = \theta \times \Gamma\left(1 + \frac{1}{\beta}\right)$$

  – Where, $\Gamma(x) = \int_0^\infty y^{x-1} e^{-y} dy$
    - $\Gamma(x) = (x-1)\Gamma(x-1)$, where $x > 0$
    - $\Gamma(x) = (x-1)!$, *for positive integer values of* x

- *Variance*

$$\sigma^2 = \theta^2\left[\Gamma\left(1 + \frac{2}{\beta}\right) - \left(\Gamma\left(1 + \frac{1}{\beta}\right)\right)^2\right]$$

- Design life

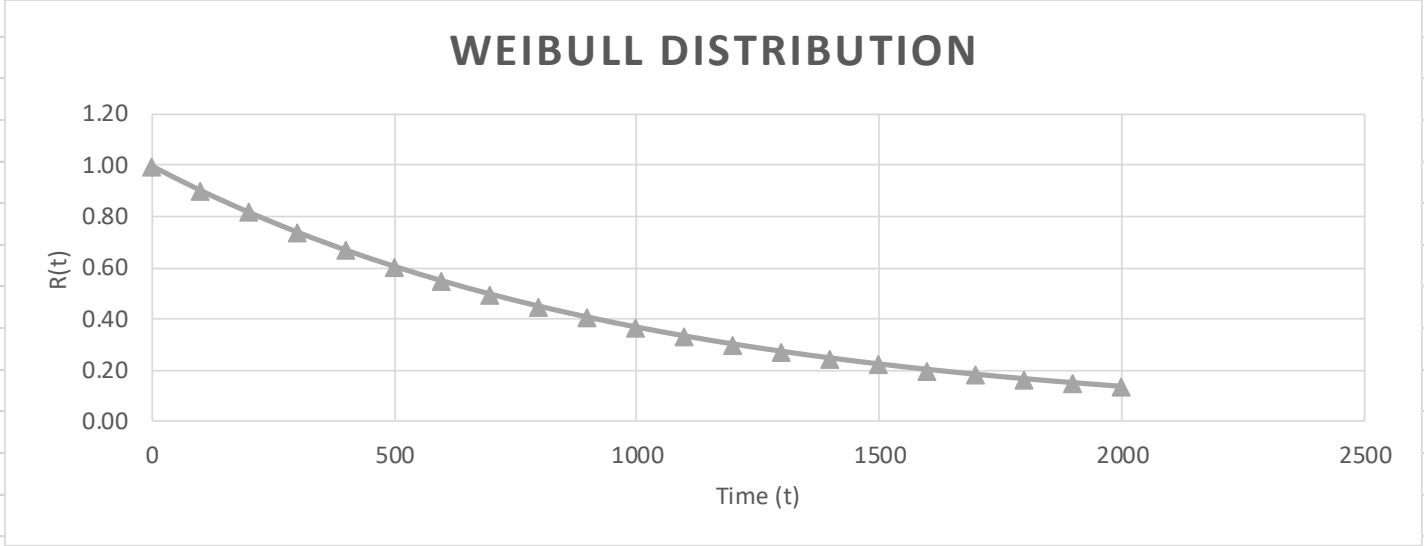
$$t_R = \theta(-\ln R)^{1/\beta}$$

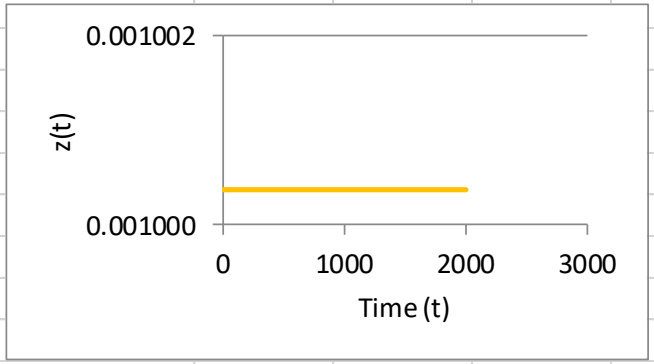
  – t_R is also called $B(1-R)100$ life.
  – For R=0.90, it is $B10$ life in which 10% of the population is expected to fail.

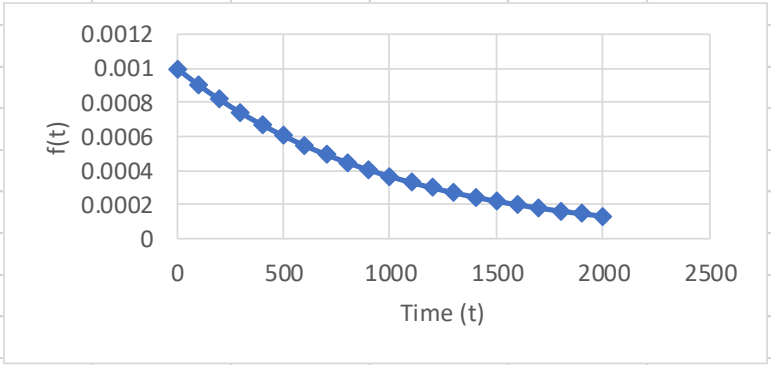
θ	1000		MTTF	1000
β	1		Var	1000000

1000

x	f(x)	F(x)	R(x)	z(x)
1	0.000999	0.00	1.00	0.001000
100	0.000905	0.10	0.90	0.001000
200	0.000819	0.18	0.82	0.001000
300	0.000741	0.26	0.74	0.001000
400	0.00067	0.33	0.67	0.001000
500	0.000607	0.39	0.61	0.001000
600	0.000549	0.45	0.55	0.001000
700	0.000497	0.50	0.50	0.001000
800	0.000449	0.55	0.45	0.001000
900	0.000407	0.59	0.41	0.001000
1000	0.000368	0.63	0.37	0.001000
1100	0.000333	0.67	0.33	0.001000
1200	0.000301	0.70	0.30	0.001000
1300	0.000273	0.73	0.27	0.001000
1400	0.000247	0.75	0.25	0.001000
1500	0.000223	0.78	0.22	0.001000
1600	0.000202	0.80	0.20	0.001000
1700	0.000183	0.82	0.18	0.001000
1800	0.000165	0.83	0.17	0.001000
1900	0.00015	0.85	0.15	0.001000
2000	0.000135	0.86	0.135335	0.001000







Limitations of Reliability Theory

- It can not predict when a system / component will actually fail .
- Reliability is 100% only if you do not use the system/component.

Maintainability Related Terms

- Maintenance
  - The combination of all technical and administrative actions, including supervision actions, intended to retain a product in, or restore it to, a state in which it can perform a required function
  - Corrective Maintenance
    - The maintenance is carried out after fault recognition and intended to put a product into a state in which it can perform required function
  - Proactive Maintenance
    - Predictive Maintenance
      - The maintenance is scheduled to be carried out cost effectively using information gathered through condition monitoring technique(s) before equipment loses its performance beyond threshold.
    - Preventive maintenance
      - The maintenance carried out at pre-determined intervals or according to prescribed criteria and intended to reduce the probability of failure or the degradation of the functioning of an item

Maintainability Related Terms …

- Maintainability
  - $H(t) = \Pr\{T \leq t\} = \int_0^t h(x)dx$
    - Where T is Time To Repair (TTR) is random variable.
    - h(x) is pdf of TTR
- Mean Time to Repair (MTTR)
  - $MTTR = E(T) = \int_0^\infty t.h(t)dt$
- Repair Rate
  - It provides an instantaneous rate of repair at time t.
  - It is conditional probability of completing repair per unit time in time interval (t, t+Δt).
  - $\mu(t) = \frac{h(t)}{1-H(t)}$

Availability Related Terms

- Average Availability
  - $A(T) = \frac{1}{T}\int_0^T A(t)dt$
  - For single unit, following exponential failure and repair distributions with parameters λ and μ.
    - $A(t) = \frac{\mu}{\mu+\lambda} + \frac{\lambda}{\mu+\lambda}e^{-(\lambda+\mu)t}$
- Interval (mission) Availability
  - $A_{t_2-t_1} = \frac{1}{t_2-t_1}\int_{t_1}^{t_2} A(t)dt$
- Inherent (Steady State) Availability
  - $A_{inh} = A_{SS} = \lim_{T\to\infty} A(T) = \frac{MTBF}{MTBF+MTTR} = \frac{MUT}{MUT+MDT_{CM}}$

Availability Related Terms …

- Achieved Availability
  - $A_a = \frac{MTBM}{MTBM + \bar{M}} = \frac{MUT}{MUT + MDT_{CM} + MDT_{PM}}$
    - where, MTBM is mean time between maintenance (all types) and
    - $\bar{M}$ is mean system downtime considering all maintenance downtimes.
- Operational Availability
  - $A_o = \frac{MTBM}{MTBM + \bar{M} + MDT} = \frac{MUT}{MUT + MDT_{CM} + MDT_{PM} + MDT_{adm}}$
    - Where, MDT is mean delay time due to maintenance and supply delay time (including administrative delays)
- Which of the above availability decreases to achieve high reliability?

Availability and Downtime

- Expected downtime in a year can be calculated as:
  - $MDT = (1 - A_{SS}) \times 365 \times 24 \times 60\ min/yr$
- For most of railway projects, availability requirements of signalling system are at least 99.99%.
- If signalling system is demanded for 99.99% availability; but vendor has provided 99.95% availability. How much is the difference in down time?

Availability (%)	Downtime (%)	Downtime (in (Hrs	Downtime (in (Min)
90%	10%	876 hrs.	52560 Min.
99%	1%	87.6 hrs.	5256 min
99.9%	0.1%	8.76 hrs.	525.6 min.
99.99%	0.01%	0.876 hrs.	52.56 min.
99.999%	0.001%	.0876 hrs.	5.256 min.

Safety Related Terms

- System safety definition as per MIL-STD-882D
  - The application of engineering and management principles, criteria and techniques to achieve acceptable mishap risk, within the constraints of operational effectiveness and suitability, time and cost, throughout all phases of system life cycle.
- Accident
  - An unintended event or series of events resulting in loss of human health or life, damage to property or environmental damage
- Hazard
  - A condition that could lead to an accident

Safe System

- A probabilistically safe system has no single point of failure, and enough redundant sensors, computers and effectors so that it is very unlikely to cause harm.
- An inherently safe system is a clever mechanical arrangement that cannot be made to cause harm
  - obviously the best arrangement, but this is not always possible.
- A fail-safe system is one that cannot cause harm when it fails.
- A fault-tolerant system can continue to operate with faults, though its operation may be degraded in some fashion.

Risk Related Terms

- Risk
  - It is the probability that a particular adverse event occur during a stated period of time or results from a particular challenge.
    - An adverse event is an occurrence that produces harm, which is loss to human being or population of human being.
  - Linear Risk

$$R = \sum_{i=1}^{n} p_i C_i$$

  - Non linear Risk

$$R = \sum_{i=1}^{n} p_i \, C_i^{\,r}$$

Involuntary Individual Risk Acceptance Limits

Fatality Risk Level per Year	Consideration
10^{-3}	This level is unacceptable to everyone. Accidents exposing hazards at this level are difficult to find. When risk approaches this level, immediate action is taken to reduce the hazard.
10^{-4}	This level is acceptable; however, individuals would like to see that public money is spent to control a hazard exposure at this level (e.g. placing traffic signs).
10^{-5}	While acceptable, the individuals in public recognize this level of risk, but generally feel comfortable about it. At this level, individuals may accept inconveniences to avoid this level of risk, such as avoiding air travel.
10^{-6}	Generally acceptable with no great concern. Individuals may be aware of the risks at this level, but feel that they cannot happen to them. They view this level of risk as “an act of God.”
Less than 10^{-6}	Acceptable. Individuals are not generally aware of this level of risk.

References

- Charles E. Ebeling (2019) “An Introduction to Reliability and Maintainability Engineering”, 3rd edition, Publisher: McGraw Hill Education.
- Technical Committee CENELEC TC9X, electrical and electronic applications in railways, “EN 50126-1: Generic RAMS process”, 2019, CENLEC

THANK YOU

Expected capability development from this lecture:

1. Knowing the RAMS terms and their definitions
2. Relationship of the terms with real world
3. Mathematical relationship among various terms
4. General understanding of nomenclature used in defining the terms.
5. Importance of knowing distributions of various terms used.
6. Capability to calculate various terms from knowledge of parameters.